power_in



File: power_in.kicad_sch

24 V input, protection, current sense, logic supply

buck_5v



File: buck_5v.kicad_sch

5 V / 15 A LED supply (LM5145)

mcu



File: mcu.kicad_sch

ESP32-S3-Zero

led_out



File: led_out.kicad_sch

LED strip outputs (8 channels, 2 blade connectors)

encoder



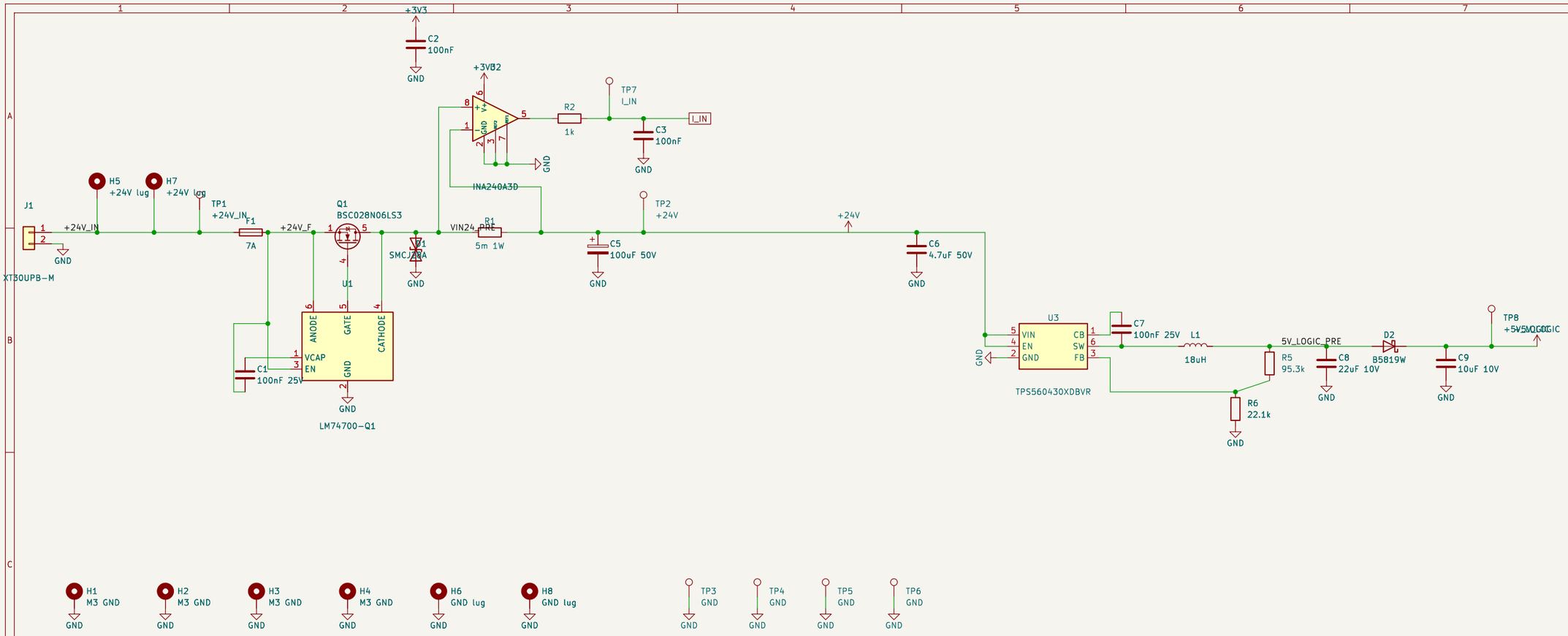
File: encoder.kicad_sch

Encoder inputs and optical link

Sheet: /
File: povd-hub.kicad_sch

Title: povd-hub: top level

Size: A4	Date: 2026-10-23	**Rev: A**
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H5/H7 (+24V_IN) and H6/H8 (GND): plated M3 holes under the cap screws, usable as ring-lug terminals instead of J1. H1-H4 (GND) mount the board.

J1 XT30: 15 A continuous. Brushes: 4 x (4 x 5 mm), 2 per polarity, ~3-5 A each copper-graphite -> 6-10 A per polarity.

F1 7 A: 2x the 3.4 A input current at the 15 A 5 V rating.

U1/Q1 ideal diode: reversed supply is blocked, ~10 mV drop at 3.4 A.

U2 INA240A3 + 5 mOhm: 0.5 V/A at I_IN (ADC1_CH0). ~6 A full scale.

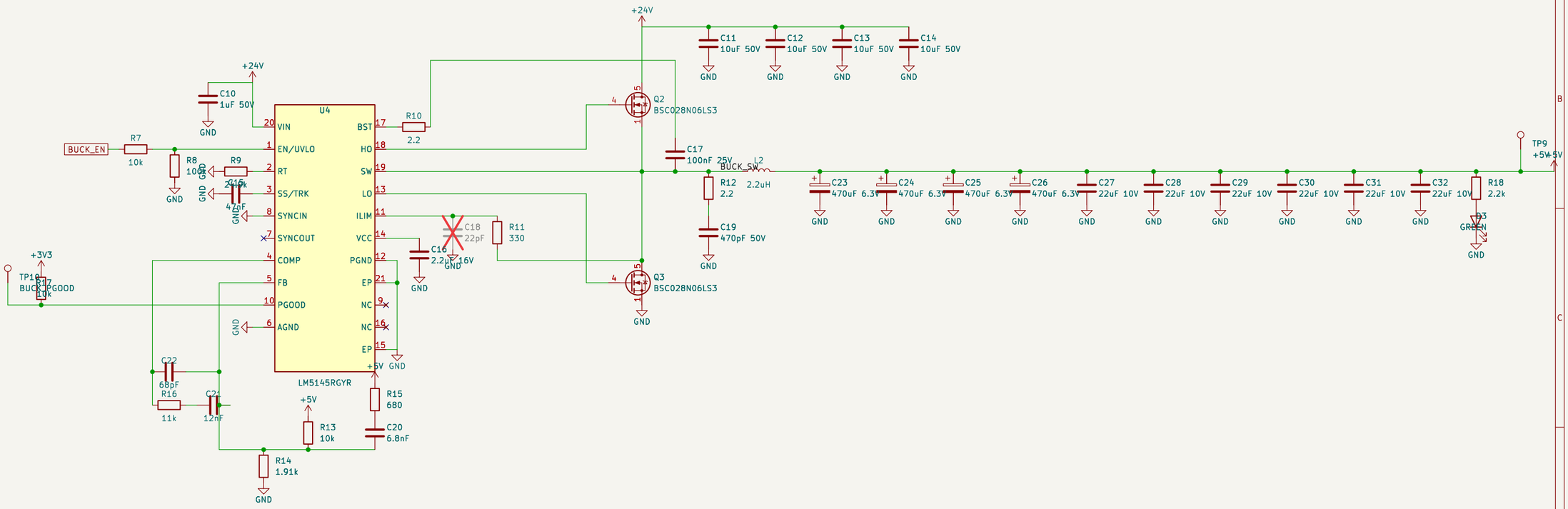
U3 TPS560430X 1.1 MHz, VOUT = 1.0 V x (1 + 95.3k/22.1k) = 5.31 V; after D2 ~5.0 V to the S3-Zero 5V pad.

See docs/hub-board.md
Rotor hub board: 24 V in, 5 V / 15 A LED rail, ESP32-S3-Zero, 8 strip outputs, encoder

Sheet: /power_in/
File: power_in.kicad_sch

Title: povd-hub: 24 V input, protection, current sense, logic supply

Size: A3	Date: 2026-10-23	**Rev: A**
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fsw 400 kHz (RT 24.9k). VOUT = 0.8 V x (1 + 10k/1.91k) = 4.99 V. Soft start ~3.8 ms (47 nF).

L2 2.2 uH: 4.5 A p–p ripple at 24 V in. Valley current limit ~18 A hot (R11 330 R, 200 uA ILIM, RDS(on) sensing on Q3).

Type–3 compensation, first pass: fc ~40 kHz with LC f0 ~2.4 kHz and 4 x 470 uF polymer.

Check R11 and R13–R16, C20–C22 in the TI LM5145 Quickstart Calculator against the final L and Cout parts.

BUCK_EN from the MCU, 100k pull–down: the LED rail stays off until firmware enables it.

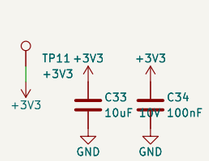
SYNCIN to GND: diode emulation at light load. Tie to VCC for forced PWM.

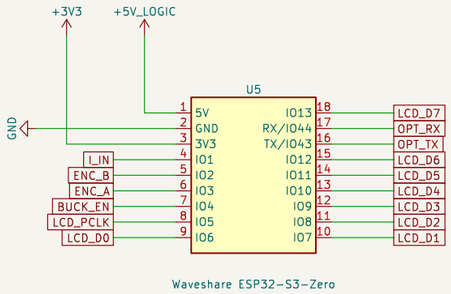
See docs/hub–board.md
Rotor hub board: 24 V in, 5 V / 15 A LED rail, ESP32–S3–Zero, 8 strip outputs, encoder

Sheet: /buck_5v/
File: buck_5v.kicad_sch

Title: povd–hub: 5 V / 15 A LED supply (LM5145)

Size: A3	Date: 2026–10–23	**Rev: A**
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Waveshare ESP32-S3-Zero

ESP32-S3-Zero on 2 x 9 headers. Only the 18 edge pins are used.

IO1 I_IN is ADC1_CH0. IO5-IO13 drive the LCD_CAM i80 bus (PCLK + D0-D7).

IO43/IO44 are UART0: the optical link. The ROM boot log appears on OPT_TX.

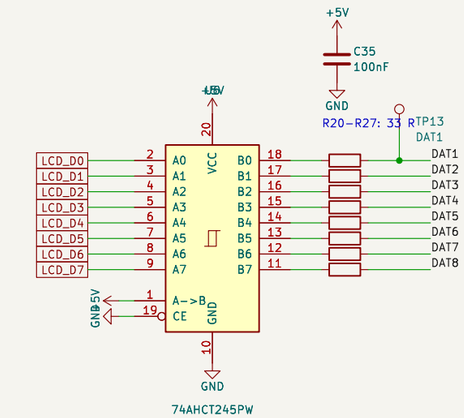
Console and flashing over the module USB-C. GPIO21 is the on-board RGB LED.

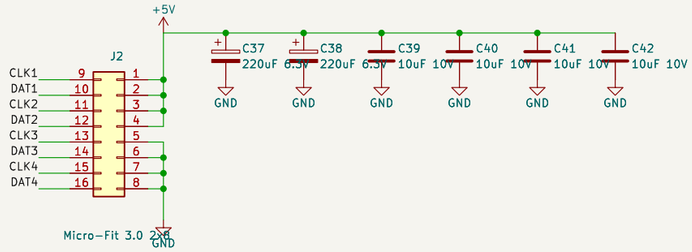
See docs/hub-board.md
Rotor hub board: 24 V in, 5 V / 15 A LED rail, ESP32-S3-Zero, 8 strip outputs, encoder

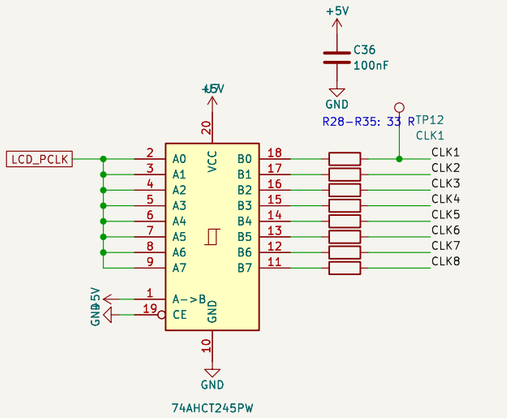
Sheet: /mcu/
File: mcu.kicad_sch

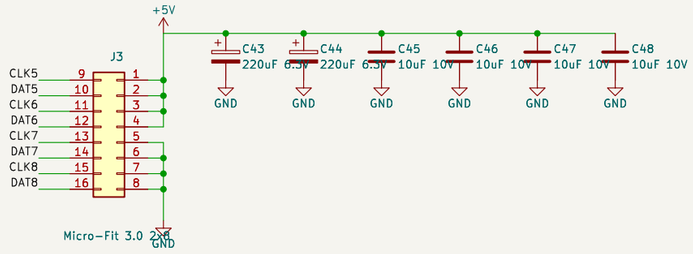
Title: povd-hub: ESP32-S3-Zero

Size: A3	Date: 2026-10-23	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/6









U6: LCD_D0–D7 -> DAT1–DAT8. U7: LCD_PCLK -> CLK1–CLK8, one buffered copy per strip.

Buffers run from +5V (LED rail): unpowered until BUCK_EN. /OE tied low.

J2 blade A = strips 1–4, J3 blade B = strips 5–8. Pins 1–4 +5V, 5–8 GND, 9–16 CLK/DAT pairs.

J2/J3 mate parallel to the rotor axis; clamp the harness at the hub so the latch carries no cable load.

See docs/hub-board.md
Rotor hub board: 24 V in, 5 V / 15 A LED rail, ESP32-S3-Zero, 8 strip outputs, encoder

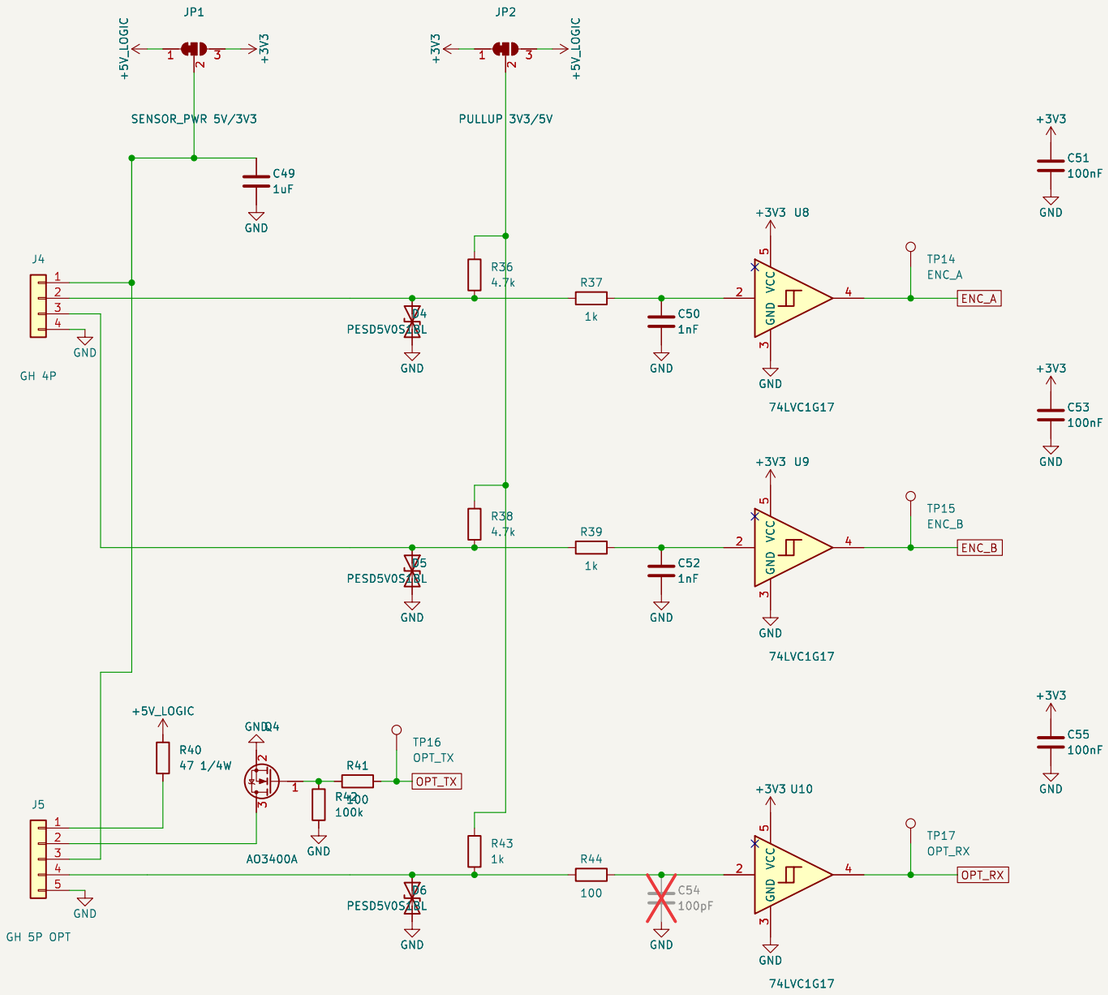
Sheet: /led_out/
File: led_out.kicad_sch

Title: povd-hub: LED strip outputs (8 channels, 2 blade connectors)

Size: A3	Date: 2026-10-23	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 5/6

JP1: sensor supply (default +5V_LOGIC)

JP2: pull-up rail (default +3V3)



J4 pins: 1 SENSOR_PWR, 2 ENC_A, 3 ENC_B, 4 GND. JP1 selects sensor supply (default 5V_LOGIC).

JP2 selects the pull-up rail (default 3V3). R36/R38 4.7k, change for the sensor type.

1k / 1 nF filter (1 us), 74LVC1G17 Schmitt at 3V3 (inputs 5 V tolerant).

Disk: 31 posts, 124 edges, 120 net counts/rev; index = high pulse ~3x longer.

J5 optical link: 1 LED anode (via R40 from +5V_LOGIC), 2 LED cathode (Q4 low side), 3 SENSOR_PWR, 4 sensor output, 5 GND.

OPT_TX is UART0 TX: set TX inversion in firmware so the LED is off when the line idles.

Receiver: R43 load to the JP2 rail, 100 R + optional 100 pF, 74LVC1G17 Schmitt -> UART0 RX.

See docs/hub-board.md
Rotor hub board: 24 V in, 5 V / 15 A LED rail, ESP32-S3-Zero, 8 strip outputs, encoder

Sheet: /encoder/
File: encoder.kicad_sch

Title: povd-hub: Encoder inputs and optical link

Size: A3	Date: 2026-10-23	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 6/6